

MATH 475: Honours Partial Differential Equations

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Abstract

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1 Introduction

Taught by Adam Oberman. Mondays and Wednesdays from 14:35 to 15:55 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% 5-6 Assignments.
- 10% Attendance and participation.
- 10% 3-5 quizzes.
- 20% midterm.
- 50% final exam.

1.1 Ways to Write a PDE

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Many of the classical equations are built from a single operator, the Laplacian

$$\Delta u = \sum_{i=1}^n \frac{\partial^2 u}{\partial x_i^2}. \quad (1)$$

Equation	Form	Type
Heat	$u_t = \Delta u, \quad t \geq 0$	parabolic
Poisson	$\Delta u = f$	elliptic (stationary)
Laplace	$\Delta u = 0$	elliptic (stationary)
Wave	$u_{tt} = \Delta u$	hyperbolic

Not every PDE fits this pattern. Nonlinear examples include Burgers' equation

$$u_t + uu_x = 0, \quad (2)$$

and the Navier-Stokes equations.

2 Potential Theory

2.1 Newton's Problem

Place a point mass M at the origin and a test mass m at $x \in \mathbb{R}^3 \setminus \{0\}$. The gravitational force on m has magnitude $GmM/\|x\|^2$ and points back toward the origin:

$$\vec{F}(x) = -\frac{GmM}{\|x\|^2} \cdot \frac{x}{\|x\|} = -\frac{GmM}{\|x\|^3} x. \quad (3)$$

2.2 Toolkit

Lemma 1 (Derivative of distance is direction). *Let $r(x) = \|x\|$ with $x = (x_1, x_2, x_3) \in \mathbb{R}^3$. Then for $x \neq 0$,*

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}, \quad \nabla r(x) = \frac{x}{r} = \hat{x}. \quad (4)$$

Proof. Start from $r^2 = x_1^2 + x_2^2 + x_3^2$ and differentiate both sides with respect to x_i :

$$2r \frac{\partial r}{\partial x_i} = 2x_i \implies \frac{\partial r}{\partial x_i} = \frac{x_i}{r}. \quad (5)$$

This requires $r \neq 0$, so the identity fails at the origin. $\square$

Remark 1. ∇r is the unit vector pointing radially outward, so $\|\nabla r\| = 1$ everywhere it is defined.

2.3 Radial Chain Rule

Proposition 1. *Let $f : (0, \infty) \rightarrow \mathbb{R}$ be differentiable and set $g(x) = f(r(x)) = f(\|x\|)$. Then for $x \neq 0$,*

$$\nabla g(x) = f'(r) \nabla r(x) = f'(r) \frac{x}{r}. \quad (6)$$

Proof. Componentwise, by the chain rule,

$$\frac{\partial g}{\partial x_i} = f'(r(x)) \frac{\partial r}{\partial x_i} = f'(r) \frac{x_i}{r}. \quad (7)$$

$\square$

So the gradient of any radial function points radially, and f' alone controls the magnitude.

Example (Gradient refresher, non-radial). *For $g(x) = x_1 x_2 + x_3^2$,*

$$\nabla g(x) = \left(\frac{\partial g}{\partial x_1}, \frac{\partial g}{\partial x_2}, \frac{\partial g}{\partial x_3} \right) = (x_2, x_1, 2x_3). \quad (8)$$

Remark 2. *The chain rule result above, $\nabla g(x) = f'(r(x)) \nabla r(x) = f'(r) \frac{x}{r}$, is the tool we use whenever a function depends only on distance from the origin.*

2.4 The Laplacian

Everything below works in $\mathbb{R}^d$ for any $d \geq 1$. Write $x = (x_1, \dots, x_d)$.

First form: sum of second partials

Definition 1 (Laplacian).

$$\Delta = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2}, \quad \Delta f(x) = \sum_{i=1}^d \frac{\partial^2 f}{\partial x_i^2}(x). \quad (9)$$

In $d = 3$ this reads

$$\Delta f(x) = \frac{\partial^2 f}{\partial x_1^2}(x) + \frac{\partial^2 f}{\partial x_2^2}(x) + \frac{\partial^2 f}{\partial x_3^2}(x). \quad (10)$$

Example. Let $f(x_1, x_2, x_3) = x_1^2 - x_2^2$. Then

$$\frac{\partial^2 f}{\partial x_1^2} = 2, \quad \frac{\partial^2 f}{\partial x_2^2} = -2, \quad \frac{\partial^2 f}{\partial x_3^2} = 0, \quad (11)$$

so $\Delta f = 2 - 2 + 0 = 0$. A function with $\Delta f = 0$ is called harmonic.

Second form: divergence of the gradient

Definition 2 (Divergence). For a vector field $\vec{v}(x) = (v^1(x), \dots, v^d(x))$, where the superscripts label components and are not powers,

$$\nabla \cdot \vec{v} = \operatorname{div} \vec{v} = \sum_{i=1}^d \frac{\partial v^i}{\partial x_i}(x). \quad (12)$$

Applying this to the gradient field $\vec{v} = \nabla f$, whose i -th component is $\partial f / \partial x_i$, gives

$$\nabla \cdot (\nabla f) = \sum_{i=1}^d \frac{\partial}{\partial x_i} \left(\frac{\partial f}{\partial x_i} \right) = \sum_{i=1}^d \frac{\partial^2 f}{\partial x_i^2} = \Delta f. \quad (13)$$

Proposition 2.

$$\Delta = \nabla \cdot \nabla = \operatorname{div} \circ \operatorname{grad}. \quad (14)$$

Remark 3. This second form is the useful one for radial functions: combining it with the radial chain rule turns Δ into an ODE in r .

2.5 Harmonic Radial Functions

Proposition 3 (Radial Laplacian). For $g(x) = f(r)$, $r = \|x\|$ in $\mathbb{R}^d$, $x \neq 0$:

$$\Delta g = f''(r) + \frac{d-1}{r} f'(r). \quad (15)$$

Solve $\Delta f = 0$ in $d = 3$ with the ansatz $f(r) = r^a$:

$$\Delta f = a(a-1)r^{a-2} + \frac{2}{r} ar^{a-1} = a(a+1)r^{a-2}, \quad (16)$$

which vanishes for $r > 0$ iff $a = 0$ or $a = -1$. So

$$g(x) = A + \frac{B}{\|x\|}. \quad (17)$$

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Remark 4. $1/\|x\|$ is the Newtonian potential: $\nabla(1/r) = -x/\|x\|^3$, the inverse-square field. Harmonic away from the origin only.

Order of PDE is the order of the highest derivative which appears in it. Example U_{xx} is of 2nd order and $U_t + UU_x$ is of 1st order. Solution of PDE (1) $u \in C^K$, where K is the order of the PDE, and u is the k -times differentiable (2) $F(u) = 0$. Example: Let $u = u(x, t)$, $F(u_t, u_x) = u_t + cu_x = 0, c \in \mathbb{R}$. A Solution, let $\Phi(z)$ be any differentiable function then define $u(x, t) = \Phi(x - ct)$ solves the PDE. $u_x = \Phi'(x - ct)$, $u_t = -c\Phi'(x - ct)$, so $u_t + cu_x = 0$. Example: $u_t = c^2 u_{xx}$, Ansatz: k is a constant, orbiting $u(x, t) = e^{-c^2 k^2 t} \sin(kx) = \exp(-c^2 k^2 t) \sin(kx)$, $u_t = -c^2 k^2 u$, $u_x = k \exp(-c^2 k^2 t) \cos(kx)$, $u_{xx} = -k^2 \exp(-c^2 k^2 t) \sin(kx) = -k^2 u$, so $u_t = c^2 u_{xx}$.

Example (wave equation): $u_{tt} = c^2 u_{xx}$, $u(x, t) = \cos(kct) \sin(kx)$, $u_{xx} = -k^2 \cos(kct) \sin(kx) = -k^2 u$, $u_{tt} = -c^2 k^2 \cos(kct) \sin(kx) = -c^2 k^2 u$, so $u_{tt} = c^2 u_{xx}$.

Classification of PDEs. Linear/Nonlinear: example $F(u, u_x, x) = 3u + u_x - x^2$, $L(u, u_x) = 3u + u_x$, $f(x) = x^2$, $F(u, u_x, x) = 0 \leftrightarrow L(u, u_x) = f(x)$ which is linear. PDE: $F(u, u_x, x) - 0$ is linear if it can be written in the form $L(u, u_x) = f(x)$, where L is linear in each term.

Similarly in higher dimensions. Sub definition, the linear PDE $L(u) = f(x)$ is homogeneous. Example $u_t + au_x x^3 = 0$ is linear $L(u_t, u_x) = u_t + au_x$, $f(x) = x^3$, it's inhomogeneous. Example 2: $u_t + uu_x = f(x)$ is nonlinear.

Principle of Superposition: Linear $L(u) = 0$, u_1, u_2 are solutions of $L(u) = 0$, then so is $u_1 + cu_2$ for all $c \in \mathbb{R}$. Check: $L(u_1 + cu_2) = L(u_1) + cL(u_2) = 0 + c \cdot 0 = 0$. Example: $u_t = c^2 u_{xx}$, $u_1(x, t) = \exp(-c^2 k^2 t) \sin(kx)$, $u_2(x, t) = \exp(-c^2 t) \sin(x)$, then $u(x, t) = a_1 u_1 + a_2 u_2$ is also a solution. ¹

3 General Solutions

Example $u(x, y)$ is a function in $\mathbb{R}^2$, solve (all), $u_x = 0$ in $\mathbb{R}^2$, solution is $u(x, y) = f(y)$ for f arbitrary. Example: $u(x, y) \in \mathbb{R}^\#$, $u_{xx} = 0$, steps: let $v(x, y) = u_x$ by previous example $v(x, y) = f(y)$ so $u_x = f(y)$ so fix one $y \Rightarrow u(x, y) = f(y) + xg(y)$. Check PDE $u(x, y) = f(y) + xg(y)$, $u_x = 0 + g(y)$, $u_{xx} = 0$.

Example: same idea $u(x, y)$, $u_{xy} = 0$, let $v(x, y) = u_x$, then $v_y = 0$ so $v(x, y) = f(x)$, so $u_x = f(x)$, fix one y and integrate in x , $u(x, y) = F(x) + g(y)$, check: $u_{xy} = F'(x)_y + g'(y)_x = 0 + 0 = 0$.

"auxiliary" conditions means boundary conditions, $\delta u = f(x)$ in domain $\Omega \subset \mathbb{R}^d$, $u(x) = g(x)$ on boundary $\partial\Omega \subset \mathbb{R}^d$.

$Lu = f$, here it's a linear PDE where L is the operator, u is the want and f is the input. $Mx = b$, $x \in \mathbb{R}^d$ vector $b \in \mathbb{R}^d$, $x = M^{-1}b$

¹80% chance of what we just wrote so far is not on the final.

- A Appendix**
- B Solutions**
- C Useful Links**